BOĞAZİÇİ UNIVERSITY

Department of Electrical and Electronics Engineering

INTERNSHIP REPORT

GNSS-Denied Aircraft Localization Using LTE Signals of Opportunity and Inertial Navigation

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Abstract

Global Navigation Satellite System (GNSS) signals may become unavailable because of obstruction, interference, jamming, or spoofing. This project investigates whether signals transmitted by LTE base stations can be used as signals of opportunity to support aircraft navigation during a GNSS outage.

Before the outage, GNSS measurements and synchronized receivers are used to estimate and store the locations of visible LTE base stations. During the outage, inter-receiver time-difference-of-arrival measurements are converted into bearing observations. These observations are used to estimate the aircraft position and aid an inertial navigation system through an extended Kalman filter.

The proposed method is evaluated using a two-dimensional MATLAB simulation containing a moving aircraft, inertial sensor errors, LTE signal processing, base-station mapping, and a simulated GNSS outage.

1 Introduction

1.1 Problem Statement

The problem considered in this project is the estimation of an aircraft's horizontal position when GNSS measurements are unavailable. The proposed system uses previously mapped LTE base stations as radio-frequency landmarks.

1.2 Project Objectives

The main objectives of the project are:

- to simulate a realistic two-dimensional aircraft trajectory;
- to simulate IMU, GNSS, and heading measurements;
- to detect and identify LTE base stations using their PCIs;
- to estimate base-station positions before the GNSS outage;
- to estimate aircraft position from LTE bearing measurements;
- to combine LTE and INS information using an EKF;
- to compare INS-only and LTE-aided navigation performance.

1.3 Scope and Limitations

The project is implemented as a two-dimensional proof-of-concept simulation. The model does not include altitude, aircraft banking, multipath propagation, non-line-of-sight propagation, clock drift, or a complete LTE network implementation.

2 Technical Background

2.1 GNSS and GNSS Outages

A Global Navigation Satellite System (GNSS) receiver estimates its position by measuring the propagation times of signals transmitted by multiple satellites. The present simulation does not model individual satellites or raw pseudoranges. Instead, the output of a completed GNSS receiver is represented as a noisy position and velocity measurement:

$$\boldsymbol{z}_{\mathrm{GNSS},k} = \begin{bmatrix} p_{E,k} \\ p_{N,k} \\ v_{E,k} \\ v_{N,k} \end{bmatrix} + \boldsymbol{n}_{\mathrm{GNSS},k}, \tag{1}$$

where $\boldsymbol{n}_{\mathrm{GNSS},k}$ represents GNSS measurement noise.

2.2 Inertial Navigation Systems

An inertial navigation system estimates motion using accelerometers and gyroscopes. The accelerometers measure specific force along the aircraft body axes, while the gyroscope measures angular rate. In this two-dimensional model, the body axes are defined as forward and left, and the navigation frame is defined by east and north.

The simulated IMU measurements are

$$\boldsymbol{a}^b_{m,k} = \boldsymbol{a}^b_k + \boldsymbol{b}_{a,k} + \boldsymbol{n}_{a,k}, \tag{2}$$

$$\omega_{m,k} = \dot{\psi}_k + b_{g,k} + n_{g,k}, \tag{3}$$

where $\boldsymbol{a}^b_{m,k}$ is the measured body-frame acceleration, $\boldsymbol{b}_{a,k}$ is the accelerometer bias, $\omega_{m,k}$ is the measured turn rate, $b_{g,k}$ is the gyroscope bias, and $\boldsymbol{n}_{a,k}$ and $n_{g,k}$ are measurement noise.

After subtracting the estimated biases, heading is propagated using

$$\hat{\psi}_k = \hat{\psi}_{k-1} + \left(\omega_{m,k} - \hat{b}_{g,k-1}\right) \Delta t. \tag{4}$$

The corrected acceleration is transformed from the aircraft body frame to the east-north navigation frame using

$$\hat{\boldsymbol{a}}^n_k = \boldsymbol{R}^n_b(\hat{\psi}_k) \left(\boldsymbol{a}^b_{m,k} - \hat{\boldsymbol{b}}_{a,k-1}\right), \tag{5}$$

where

$$\boldsymbol{R}^n_b(\psi) = \begin{bmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{bmatrix}. \tag{6}$$

Velocity and position are then propagated by integration:

$$\hat{\boldsymbol{v}}_k = \hat{\boldsymbol{v}}_{k-1} + \hat{\boldsymbol{a}}^n_k \Delta t, \tag{7}$$

$$\hat{\boldsymbol{p}}_k = \hat{\boldsymbol{p}}_{k-1} + \hat{\boldsymbol{v}}_{k-1}\Delta t + \frac{1}{2}\hat{\boldsymbol{a}}^n_k \Delta t^2. \tag{8}$$

A simplified horizontal navigation state is

$$\boldsymbol{x}_k = \begin{bmatrix} p_{E,k} & p_{N,k} & v_{E,k} & v_{N,k} & \psi_k & b_{a,f,k} & b_{a,l,k} & b_{g,k} \end{bmatrix}^\mathsf{T}, \tag{9}$$

where $b_{a,f}$ and $b_{a,l}$ are the forward- and left-axis accelerometer biases. Because acceleration and angular-rate errors are integrated, even small IMU biases cause the INS position error to grow with time. External measurements are therefore required to correct the INS periodically.

2.3 LTE Signals of Opportunity

Signals of opportunity are radio signals that were created for a purpose other than navigation but can also provide useful positioning information. LTE base stations continuously transmit downlink synchronization signals with known structures. A passive receiver can detect these signals without transmitting data or requesting positioning support from the cellular network.

In this project, LTE signals serve two purposes. First, the synchronization sequences provide a physical cell identity that is needed to distinguish and identify base stations. Second, the difference between their arrival times at synchronized aircraft-mounted receivers provides information about the signal arrival direction. The base-station locations are estimated while GNSS is available and are then used as radio-frequency landmarks during the GNSS outage.

2.4 PSS, SSS, and Physical Cell Identity

Each LTE physical cell identity (PCI) consists of two components. The Primary Synchronization Signal (PSS) identifies $N_{\mathrm{ID}}^{(2)}$, which can take the values 0, 1, or 2. The Secondary Synchronization Signal (SSS) identifies $N_{\mathrm{ID}}^{(1)}$, which can take values from 0 to 167.

The receiver detects a synchronization sequence by correlating the received samples with known candidate sequences. For PSS candidate q and timing offset ℓ, a correlation metric can be written as

$$C_q[\ell] = \left| \sum_{n=0}^{L-1} y[n+\ell] p_q^*[n] \right|^2 , \tag{10}$$

where $y[n]$ is the received signal, $p_q[n]$ is a candidate PSS sequence, L is the sequence length, and $(\cdot)^*$ denotes complex conjugation. The detected sequence and timing are obtained from the largest correlation value:

$$\left(\hat{N}_{\mathrm{ID}}^{(2)}, \hat{\ell}\right) = \arg\max_{q,\ell} C_q[\ell]. \tag{11}$$

After PSS detection, the SSS is similarly compared with the candidate SSS sequences to estimate $N_{\mathrm{ID}}^{(1)}$. The PCI is then calculated as

$$N_{\mathrm{ID}}^{\mathrm{cell}} = 3N_{\mathrm{ID}}^{(1)} + N_{\mathrm{ID}}^{(2)}. \tag{12}$$

This produces one of 504 possible PCI values, ranging from 0 to 503. In the proposed system, the detected PCI is used to associate each received signal with the corresponding mapped base station.

2.5 Time Difference of Arrival

Let $\boldsymbol{s}$ denote a base-station position and let $\boldsymbol{r}_i$ and $\boldsymbol{r}_1$ denote the positions of receiver i and the reference receiver. The exact arrival-time difference is

$$\Delta\tau_i = \frac{\|\boldsymbol{s} - \boldsymbol{r}_i\| - \|\boldsymbol{s} - \boldsymbol{r}_1\|}{c} \tag{13}$$

where c is the speed of light and $n_{\tau,i}$ is TDOA measurement noise.

Because the distance to an LTE base station is much greater than the distance between the receivers, the incoming signal can be approximated as a plane wave. Let

$$\boldsymbol{b}_i = \boldsymbol{r}_i - \boldsymbol{r}_1 \tag{14}$$

be the receiver baseline and let $\boldsymbol{d}$ be a unit vector pointing from the aircraft toward the base station. The far-field model becomes

$$\Delta\tau_i \approx -\frac{\boldsymbol{b}_i^\top \boldsymbol{d}}{c} \tag{15}$$

The minus sign results from defining $\boldsymbol{d}$ as pointing from the aircraft toward the transmitter. If the direction vector is instead defined in the direction of wave propagation, the sign is reversed.

For multiple receiver baselines, the equations can be written as

$$\Delta\boldsymbol{\tau} = -\frac{1}{c}\boldsymbol{B}\boldsymbol{d} \tag{16}$$

where each row of $\boldsymbol{B}$ contains one receiver baseline. A least-squares direction estimate is continued with normalization. The corresponding global bearing angle is

$$\hat{\theta} = \operatorname{atan2}\left(\hat{d}_N, \hat{d}_E\right). \tag{17}$$

When the direction is initially estimated in aircraft body coordinates, the measured aircraft heading is used to rotate it into the east-north coordinate frame.

2.6 Bearing-Based Positioning

After a base station has been mapped, its known position and measured bearing define a line on which the aircraft should lie. Let $\boldsymbol{s}_j$ be the position of base station j, and let $\hat{\boldsymbol{d}}_j$ be the unit vector pointing from the aircraft toward that base station. A vector perpendicular to this bearing is

$$\boldsymbol{n}_j = \begin{bmatrix} -\hat{d}_{N,j} \\ \hat{d}_{E,j} \end{bmatrix}. \tag{18}$$

The aircraft position $\boldsymbol{p}$ must satisfy the bearing-line constraint

$$\boldsymbol{n}_j^\top \boldsymbol{p} = \boldsymbol{n}_j^\top \boldsymbol{s}_j. \tag{19}$$

Stacking the constraints from all visible base stations gives

$$\boldsymbol{A}\hat{\boldsymbol{p}}_{\mathrm{ref}} = \boldsymbol{q}, \tag{20}$$

where row j of $\boldsymbol{A}$ is $\boldsymbol{n}_j^\top$, and element j of $\boldsymbol{q}$ is $\boldsymbol{n}_j^\top \boldsymbol{s}_j$.

Finally, the reference-receiver position is converted to the aircraft-centre position:

$$\hat{\boldsymbol{p}}_{\mathrm{LTE}} = \hat{\boldsymbol{p}}_{\mathrm{ref}} - \boldsymbol{R}_b^n(z_\psi)\boldsymbol{r}_{\mathrm{ref}}^b. \tag{21}$$

The implementation requires at least three base stations. If their bearing lines are nearly parallel, the geometry becomes poorly conditioned and small bearing errors can cause large position errors.

2.7 Extended Kalman Filter

The extended Kalman filter predicts the navigation state using IMU measurements and corrects it using heading, GNSS, and LTE measurements. The prediction model is nonlinear because body-frame acceleration is rotated using the estimated heading.

The state and covariance predictions are

$$\hat{\boldsymbol{x}}_k^- = f\left(\hat{\boldsymbol{x}}_{k-1}^+, \boldsymbol{u}_k\right), \tag{22}$$
$$\boldsymbol{P}_k^- = \boldsymbol{F}_k \boldsymbol{P}_{k-1}^+ \boldsymbol{F}_k^\top + \boldsymbol{Q}_k, \tag{23}$$

where $\boldsymbol{u}_k$ contains the accelerometer and gyroscope measurements, $\boldsymbol{F}_k$ is the Jacobian of the nonlinear process model, and $\boldsymbol{Q}_k$ represents IMU noise and bias random walks.

Three measurement types are used:

$$z_{\psi,k} = \psi_k + n_{\psi,k}, \tag{24}$$
$$\boldsymbol{z}_{\mathrm{GNSS},k} = \begin{bmatrix} p_E & p_N & v_E & v_N \end{bmatrix}^\top + \boldsymbol{n}_{\mathrm{GNSS},k}, \tag{25}$$
$$\boldsymbol{z}_{\mathrm{LTE},k} = \begin{bmatrix} p_E & p_N \end{bmatrix}^\top + \boldsymbol{n}_{\mathrm{LTE},k}. \tag{26}$$

For each available measurement, the innovation, Kalman gain, and corrected state are calculated as

$$\boldsymbol{y}_k = \boldsymbol{z}_k - \boldsymbol{H}_k \hat{\boldsymbol{x}}_k^-, \tag{27}$$
$$\boldsymbol{S}_k = \boldsymbol{H}_k \boldsymbol{P}_k^- \boldsymbol{H}_k^\top + \boldsymbol{R}_k, \tag{28}$$
$$\boldsymbol{K}_k = \boldsymbol{P}_k^- \boldsymbol{H}_k^\top \boldsymbol{S}_k^{-1}, \tag{29}$$
$$\hat{\boldsymbol{x}}_k^+ = \hat{\boldsymbol{x}}_k^- + \boldsymbol{K}_k \boldsymbol{y}_k. \tag{30}$$

The covariance is updated using the Joseph form:

$$\boldsymbol{P}_k^+ = \left(\boldsymbol{I} - \boldsymbol{K}_k \boldsymbol{H}_k\right) \boldsymbol{P}_k^- \left(\boldsymbol{I} - \boldsymbol{K}_k \boldsymbol{H}_k\right)^\top + \boldsymbol{K}_k \boldsymbol{R}_k \boldsymbol{K}_k^\top. \tag{31}$$

Heading measurements are applied throughout the simulation. Before the outage, GNSS position and velocity measurements also correct the filter. During the outage, the baseline solution continues using the IMU and heading sensor, while the LTE-aided solution additionally uses LTE position fixes.

3 Proposed System and Methodology

3.1 Overall System Architecture

The proposed system has two operating stages. Before the GNSS outage, GNSS position, heading, and inter-receiver TDoA measurements are used to estimate the positions of the detected LTE base stations. The resulting base-station map is then frozen.

During the GNSS outage, TDoA measurements are converted into bearings toward the mapped base stations. Multiple bearings are combined to produce an LTE position fix, which is used to correct the inertial navigation solution through an extended Kalman filter.

The complete processing sequence is:

1. generate the aircraft trajectory and sensor measurements;
2. detect each LTE physical cell identity;
3. associate each TDOA measurement with a base station;
4. map the base stations while GNSS is available;

5. calculate LTE position fixes during the outage; and
6. fuse the LTE fixes with the INS.

3.2 Coordinate Systems and Aircraft Trajectory Generation

A two-dimensional east-north coordinate system is used. Aircraft heading is measured counter-clockwise from east. The aircraft body frame consists of a forward axis and a left axis.

The aircraft trajectory is generated using commanded forward acceleration a_k and turn rate ω_k:

$$v_{k+1} = v_k + a_k \Delta t, \tag{32}$$

$$\psi_{k+1} = \psi_k + \omega_k \Delta t. \tag{33}$$

Position is updated using the average speed and heading over each time interval:

$$\boldsymbol{p}_{k+1} = \boldsymbol{p}_k + \bar{v}_k \begin{bmatrix} \cos \bar{\psi}_k \\ \sin \bar{\psi}_k \end{bmatrix} \Delta t. \tag{34}$$

Three synchronized receivers are placed at known positions in the aircraft body frame. Their global positions are calculated as

$$\boldsymbol{r}^n_{i,k} = \boldsymbol{p}_k + \boldsymbol{R}^n_b(\psi_k)\boldsymbol{r}^b_i, \tag{35}$$

where $\boldsymbol{r}^b_i$ is the body-frame position of receiver i.

3.3 Sensor Simulation

The aircraft is equipped with a simulated two-axis accelerometer, vertical-axis gyroscope, GNSS receiver, and independent heading sensor. The IMU measurement model follows the bias-and-noise model described in Section 2.2. GNSS provides noisy horizontal position and velocity measurements at 1 Hz until the beginning of the simulated outage, following the model described in Section 2.1. The heading sensor remains available throughout the simulation, providing measurements corrupted by zero-mean Gaussian noise. These measurements are used both by the EKF and to transform LTE arrival directions from the aircraft body frame to the east-north frame.

3.4 LTE Waveform Generation and PCI Detection

PSS and SSS reference signals are generated using an FFT size of 2048, a sampling rate of 30.72 MHz, and the central 62 synchronization subcarriers. A cyclic prefix is added to each OFDM symbol.

For each base station, the corresponding SSS and PSS symbols are inserted into a received buffer and complex Gaussian noise is added. The receiver searches over the three possible PSS sequences using correlation. The detected PSS identifies $N^{(2)}_{\mathrm{ID}}$ and the synchronization timing.

The preceding SSS symbol is then OFDM-demodulated and compared with all 168 possible SSS candidates. The PCI is calculated from the detected values as

$$N^{\mathrm{cell}}_{\mathrm{ID}} = 3N^{(1)}_{\mathrm{ID}} + N^{(2)}_{\mathrm{ID}}. \tag{36}$$

The detected PCI is matched with a local base-station catalogue. The current simulation assigns a unique PCI to each base station.

Each base station is processed separately during the IQ experiment. PCI detection is performed once per station, and the result is reused at all cellular observation times. The TDOA measurements are generated separately from the receiver-to-base-station ranges and are not extracted from the simulated IQ waveforms.

3.5 Pre-Outage Base-Station Mapping

Ten observation times are selected across the period before the GNSS outage. At each observation, the measured GNSS position and heading are used to calculate the global receiver positions.

The associated TDOAs are first converted into approximate bearings. Bearing lines from different aircraft positions are then intersected to obtain an initial estimate of each base-station position.

The initial estimate is refined using the exact range-difference model. For base station j, the final estimate is found from

$$\hat{\boldsymbol{s}}_j = \arg\min_{\boldsymbol{s}_j} \sum_m \sum_i \left[\|\boldsymbol{s}_j - \boldsymbol{r}_{m,i}\| - \|\boldsymbol{s}_j - \boldsymbol{r}_{m,1}\| - c\Delta\tau_{m,i,j}\right]^2, \tag{37}$$

where m denotes the mapping observation and receiver 1 is the reference receiver. The optimization is performed separately for each base station. Simulation-truth positions are used only to evaluate the mapping accuracy. After mapping, the estimated base-station positions are frozen.

3.6 GNSS-Outage Position Estimation

During the outage, the TDOAs associated with each mapped base station are converted into a body-frame arrival direction:

$$\tilde{\boldsymbol{d}}_b = (-\boldsymbol{B}_b)^{\dagger}\, c\Delta\boldsymbol{\tau}, \tag{38}$$

where $\boldsymbol{B}_b$ contains the receiver baselines and $(\cdot)^{\dagger}$ denotes the pseudoinverse. The direction is normalized and rotated into the global frame:

$$\hat{\boldsymbol{d}}_n = \boldsymbol{R}_b^n(z_\psi)\frac{\tilde{\boldsymbol{d}}_b}{\left\|\tilde{\boldsymbol{d}}_b\right\|}. \tag{39}$$

The current experiment uses perfect simulated TDOAs. However, a TDOA standard deviation of 0.5 ns is assumed when calculating the bearing and position uncertainty.

A position fix requires bearings from at least three mapped base stations. The bearing lines are combined using weighted least squares:

$$\hat{\boldsymbol{p}}_{\text{ref}} = \left(\boldsymbol{A}^{\mathsf{T}}\boldsymbol{W}\boldsymbol{A}\right)^{-1}\boldsymbol{A}^{\mathsf{T}}\boldsymbol{W}\boldsymbol{q}. \tag{40}$$

The weighting matrix accounts for bearing uncertainty, heading uncertainty, and base-station mapping uncertainty. The weights are updated twice because the uncertainty depends on the estimated distance to each base station.

The resulting position corresponds to the reference receiver. It is converted to the aircraft-centre position using

$$\hat{\boldsymbol{p}}_{\text{LTE}} = \hat{\boldsymbol{p}}_{\text{ref}} - \boldsymbol{R}_b^n(z_\psi)\boldsymbol{r}_{\text{ref}}^b. \tag{41}$$

Table 1: Primary simulation parameters.

Parameter	**Value**	**Unit**
Simulation duration	1000	s
Simulation time step	0.1	s
IMU and heading rate	10	Hz
GNSS sampling rate	1	Hz
Cellular sampling rate	1	Hz
GNSS outage start	100	s
Accelerometer bias	$0.015, -0.010$	m/s^2
Accelerometer noise	0.03	m/s^2
Gyroscope bias	0.02	deg/s
Gyroscope noise	0.05	deg/s
GNSS position noise	2.0	m
GNSS velocity noise	0.10	m/s
Heading noise	0.10	deg

A position fix is discarded if the bearing geometry is rank deficient or poorly conditioned.

3.7 LTE-Aided INS Fusion

Two navigation solutions are run for comparison. Both solutions use IMU prediction, heading updates, and GNSS position and velocity updates before the outage. During the outage, the baseline solution continues using the IMU and heading sensor, while the LTE-aided solution additionally receives LTE position updates.

The LTE measurement is

$$\boldsymbol{z}_{\mathrm{LTE},k} = \hat{\boldsymbol{p}}_{\mathrm{LTE},k}, \tag{42}$$

and its measurement covariance is obtained from the bearing-position calculation.

The final comparison therefore evaluates a heading-aided INS baseline against the same navigation system with additional LTE position aiding.

4 Simulation Setup

4.1 Primary Simulation Parameters

The primary simulation parameters are summarized in table 1. Independent random seeds are used for the IMU, GNSS, heading, TDOA, and LTE IQ simulations to ensure repeatable results.

The current nominal experiment uses perfect simulated TDOA measurements. However, a standard deviation of 0.5 ns is assumed when calculating the bearing and LTE position covariances.

4.2 Receiver Configuration

Three synchronized receivers are placed in a non-collinear configuration on the aircraft. Their body-frame positions are given in table 2.

Receiver 1 is used as the TDOA reference. The two receiver baselines have equal lengths of approximately 10.77 m, giving a maximum theoretical TDOA magnitude of approximately 35.93 ns.

The baselines are non-collinear and therefore form a rank-two geometry. This is necessary because a collinear receiver arrangement cannot uniquely estimate a two-dimensional arrival direction.

Table 2: Receiver positions relative to the aircraft centre.

Receiver	Forward (m)	Left (m)
1 (reference)	5	0
2	-5	-4
3	-5	4

Table 3: Nominal LTE base-station configuration.

Station	East (km)	North (km)	PCI
1	8.0	-0.5	10
2	11.0	3.0	71
3	4.0	4.0	184
4	6.0	7.0	307
5	12.0	8.0	402
6	9.5	10.0	499
7	14.1	5.0	25
8	13.5	7.0	138

4.3 Base-Station Configuration

Eight LTE base stations are placed at fixed east-north positions. Their locations and physical cell identities are listed in table 3.

The stations are distributed across the region containing the aircraft trajectory to provide bearings from different directions. The simulation assumes that every station is continuously observable; signal blockage, limited coverage, and distance-dependent detection are not modelled.

Each station is assigned a unique PCI within the simulated local catalogue. In a real network, PCI values may be reused and would not always uniquely identify a base station.

The aircraft trajectory, base-station positions, and aircraft positions for the outage start at 100 seconds is illustrated in figure 1.

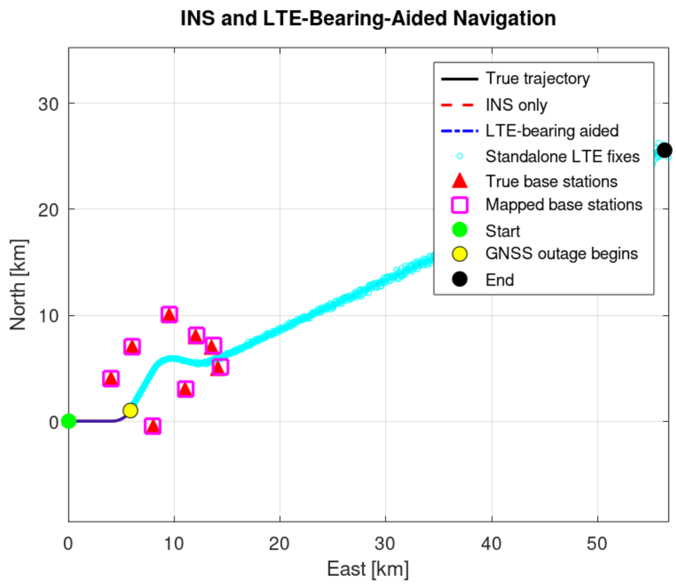


Figure 1: Aircraft trajectory and LTE base-station geometry

4.4 Test Scenarios

A $2 \times 3 \times 2$ full-factorial experiment was performed to investigate the effects of GNSS outage timing, heading uncertainty, and the number of available base stations.

The considered factors were:

- two GNSS outage starting times: 100 s and 400 s;
- three heading-noise standard deviations: 0.1°, 0.2°, and 0.5°; and
- two base-station counts: 4 and 8.

All combinations were tested, resulting in 12 configurations. All other simulation parameters and random seeds were kept constant.

4.5 Evaluation Metrics

The horizontal position error at time k is

$$e_h(k) = \sqrt{\left(\hat{p}_E(k) - p_E(k)\right)^2 + \left(\hat{p}_N(k) - p_N(k)\right)^2}. \tag{43}$$

For a set of time indices $\mathcal{K}$, the horizontal RMSE is

$$\mathrm{RMSE}_{\mathcal{K}} = \sqrt{\frac{1}{|\mathcal{K}|} \sum_{k \in \mathcal{K}} e_h^2(k)}. \tag{44}$$

The base-station mapping error is calculated as

$$e_{\mathrm{map},j} = \left\| \hat{\boldsymbol{s}}_j - \boldsymbol{s}_j \right\|. \tag{45}$$

The relative improvement in outage RMSE is

$$I_{\mathrm{RMSE}} = 100 \frac{\mathrm{RMSE}_{\mathrm{baseline}} - \mathrm{RMSE}_{\mathrm{LTE}}}{\mathrm{RMSE}_{\mathrm{baseline}}}. \tag{46}$$

A positive value indicates that LTE aiding improved the result, whereas a negative value indicates that it worsened the result.

The reported metrics include the pre-outage and outage RMSE, maximum outage error, final error, standalone LTE error, PCI detection accuracy, base-station mapping error, and the number of accepted and rejected LTE updates.

5 Results

The experiment contains 12 configurations formed from two outage starting times, three heading-noise levels, and two base-station counts. Numerical comparisons are presented in tables. The complete horizontal-error histories are grouped into six-panel figures for the 100 s and 400 s outage cases.

5.1 LTE Cell-Identification Results

The PCI-detection results are independent of heading noise and outage starting time. For the eight-station configuration, all eight PCIs were detected correctly. The 8008 cellular measurements were also associated correctly, giving an association accuracy of 100 %.

For the four-station configuration, all four PCIs were detected correctly. Its 4004 cellular measurements were associated correctly, also giving an association accuracy of 100 %.

Table 4: Summary of LTE PCI detection and association.

BS count	Detected	Correct	Associations	Accuracy
4	4	4	4004	100 %
8	8	8	8008	100 %

Table 5: PCI-detection results for the eight-station configuration.

Station	True PCI	Detected PCI
1	10	10
2	71	71
3	184	184
4	307	307
5	402	402
6	499	499
7	25	25
8	138	138

The individual PCI results for the eight-station configuration are shown in table 5.

These results verify PCI generation, correlation, and catalogue association under the simplified single-station IQ experiment. They do not represent detection from a composite multi-cell waveform.

5.2 Base-Station Mapping Results

The base-station map depends on the outage starting time and heading noise because these parameters determine the available mapping interval and the heading measurements used to construct the receiver positions.

As a representative result, all eight stations produced usable landmarks with a heading-noise standard deviation of 0.1° and an outage starting at 100 s. The mapping results are given in table 6.

If the four-station configuration contains stations 1–4, its mean mapping error for the same heading and outage settings is 14.82 m.

The estimated maps are frozen at the beginning of each outage. Simulation-truth base-station positions are used only to calculate mapping errors and are not supplied to the navigation algorithm.

5.3 Pre-Outage EKF Estimates

The reported bias values are the EKF's posterior estimates at the final sample immediately before the GNSS outage. They represent the bias estimates available at the beginning of GNSS-denied navigation, rather than the final estimates at the end of the simulation.

The true simulated biases are

$$b_{a,f} = 0.015\,\mathrm{m/s^2}, \qquad b_{a,l} = -0.010\,\mathrm{m/s^2}, \qquad b_g = 0.020\,\mathrm{deg/s}.$$

Because LTE updates are not applied before the outage, the base-station count does not affect these estimates. The six distinct pre-outage bias results are shown in table 7.

Table 6: Base-station mapping results for eight base stations, $\sigma_\psi = 0.1°$, and an outage starting at 100 s.

BS	True E	True N (m)	Est. E	Est. N	Error (m)
1	8000	-500	7987.12	-493.93	14.24
2	11000	3000	11028.18	3011.10	30.29
3	4000	4000	3998.30	4004.85	5.14
4	6000	7000	6000.40	7009.59	9.60
5	12000	8000	12027.72	8028.95	40.08
6	9500	10000	9507.56	10019.26	20.69
7	14100	5000	14168.30	5032.38	75.59
8	13500	7000	13550.38	7036.69	62.32
				Mean position error	**32.24**

Table 7: EKF bias estimates immediately before the GNSS outage.

Outage start (s)	$\boldsymbol{\sigma_\psi}$ **(deg)**	$\hat{b}_{a,f}$ (m/s^2)	$\hat{b}_{a,l}$	$\hat{b}_g$ (deg/s)
100	0.1	0.01663	-0.00988	0.02306
100	0.2	0.01661	-0.00986	0.02318
100	0.5	0.01641	-0.00982	0.02282
400	0.1	0.01377	-0.00973	0.02225
400	0.2	0.01377	-0.00973	0.02257
400	0.5	0.01376	-0.00973	0.02274

5.4 Heading-Aided INS Baseline

The baseline solution uses IMU propagation and heading updates during the outage, but it receives no external position or velocity measurements. Because the baseline does not use LTE, its result is independent of the number of base stations.

The later outage produced considerably smaller baseline errors. However, the two outage cases have different durations and different pre-outage convergence periods, so this difference cannot be attributed only to the aircraft's position at outage onset.

5.5 Results for the 100 s Outage

The outage beginning at 100 s lasted 900 s. Its standalone LTE and LTE-aided navigation results are presented in table 9.

LTE aiding reduced the outage RMSE in five of the six configurations. The largest improvement was 91.68 %, obtained with eight base stations and 0.2° heading noise. The only deterioration occurred with four base stations and 0.5° heading noise, for which the outage RMSE increased by 25.40 %.

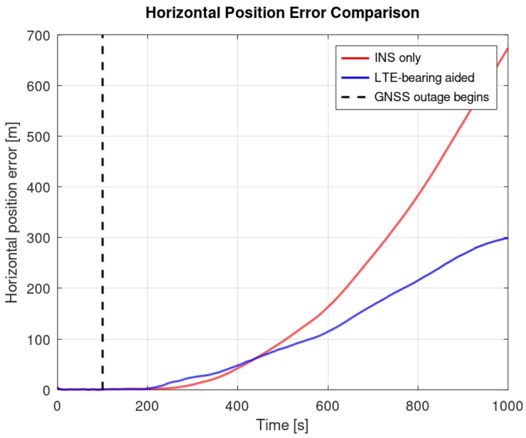
The horizontal-error histories for the six configurations are shown in figure 2. Each panel compares the heading-aided INS baseline with the LTE-aided INS solution.

5.6 Results for the 400 s Outage

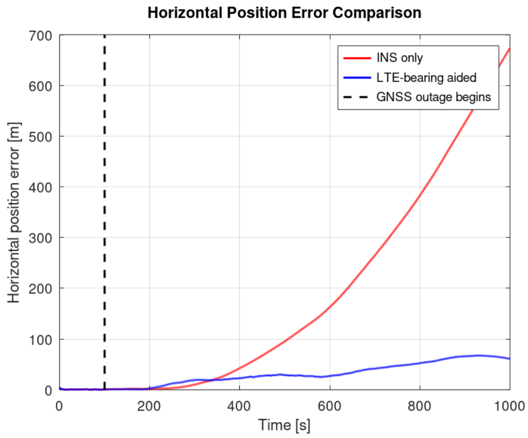
The outage beginning at 400 s lasted 600 s. The corresponding results are presented in table 10.

LTE aiding reduced the outage RMSE in all six configurations. The improvement ranged from 3.22 % to 24.35 %. The largest improvement was obtained with eight base stations and 0.5° heading noise.

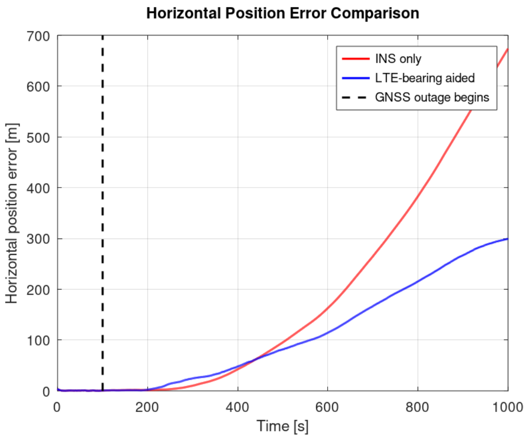
The corresponding horizontal-error histories are shown in figure 3.



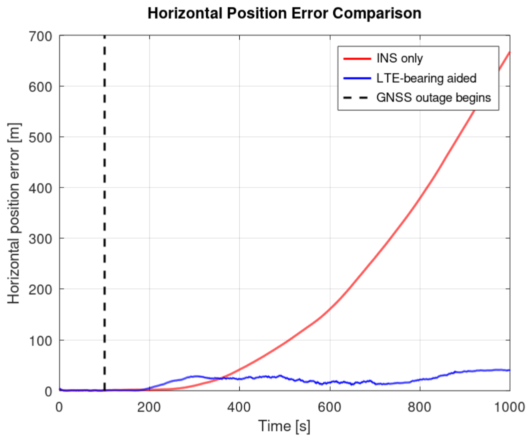
(a) $\sigma_\psi = 0.1°$, 4 base stations.



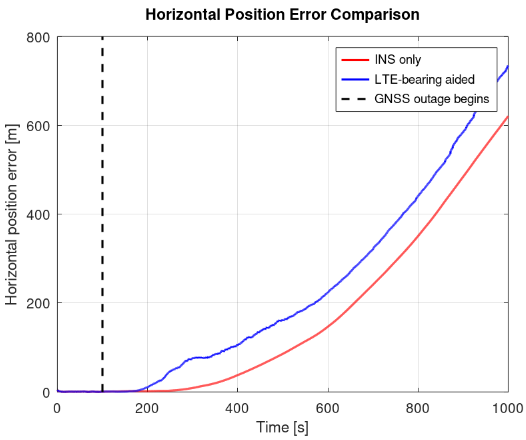
(b) $\sigma_\psi = 0.1°$, 8 base stations.



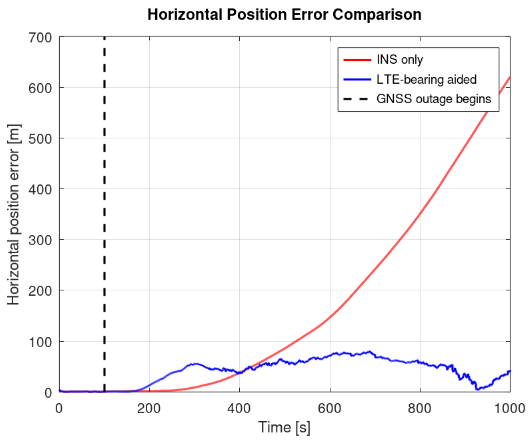
(c) $\sigma_\psi = 0.2°$, 4 base stations.



(d) $\sigma_\psi = 0.2°$, 8 base stations.

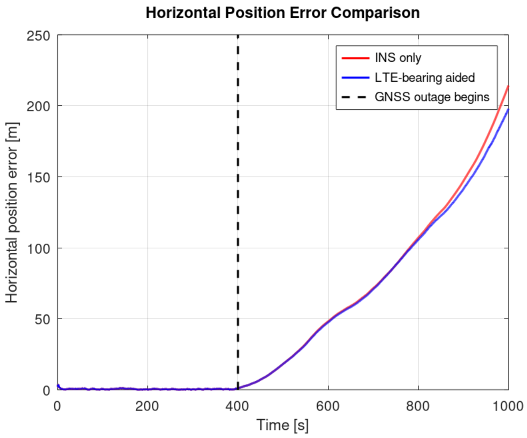


(e) $\sigma_\psi = 0.5°$, 4 base stations.

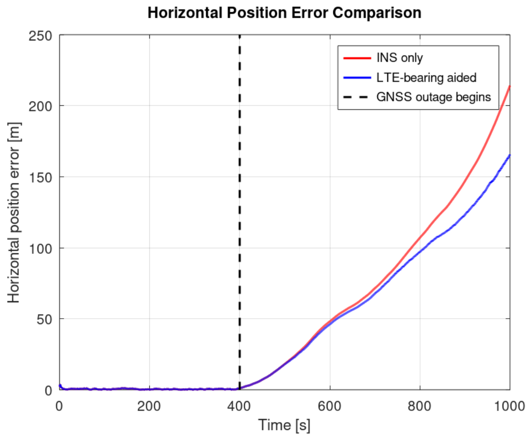


(f) $\sigma_\psi = 0.5°$, 8 base stations.

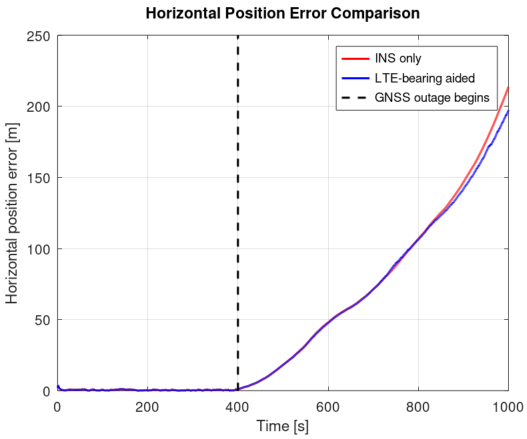
Figure 2: Horizontal position error for the six configurations with the GNSS outage starting at 100 s. The vertical dashed line indicates the beginning of the outage.



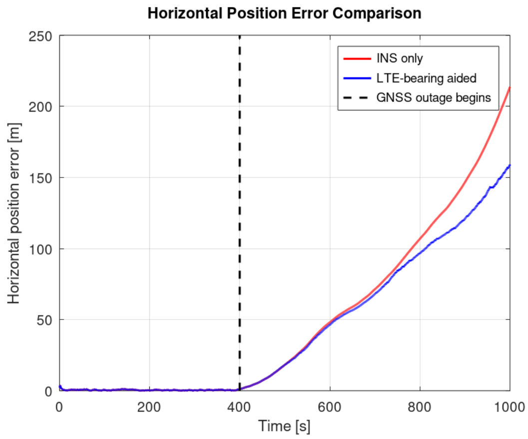
(a) $\sigma_\psi = 0.1°$, 4 base stations.



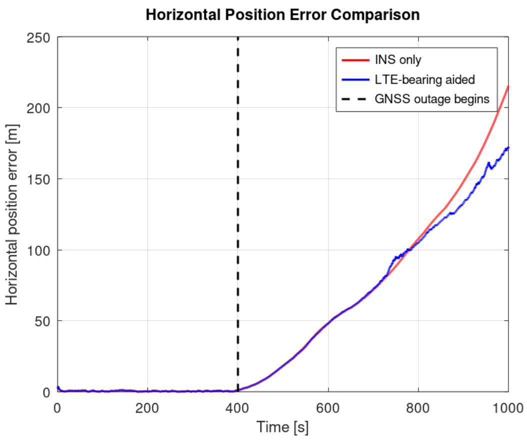
(b) $\sigma_\psi = 0.1°$, 8 base stations.



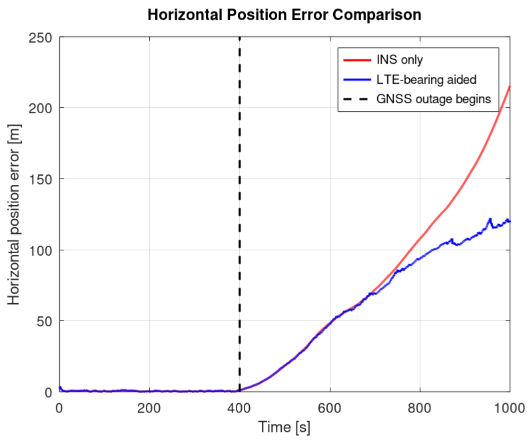
(c) $\sigma_\psi = 0.2°$, 4 base stations.



(d) $\sigma_\psi = 0.2°$, 8 base stations.



(e) $\sigma_\psi = 0.5°$, 4 base stations.



(f) $\sigma_\psi = 0.5°$, 8 base stations.

Figure 3: Horizontal position error for the six configurations with the GNSS outage starting at 400 s. The vertical dashed line indicates the beginning of the outage.

Table 8: Heading-aided INS baseline results.

Outage start (s)	σ_ψ (deg)	Pre-outage RMSE (m)	Outage RMSE (m)	Maximum error (m)	Final error (m)
100	0.1	0.80	287.21	673.60	673.60
100	0.2	0.80	284.23	667.46	667.46
100	0.5	0.80	263.60	620.66	620.66
400	0.1	0.60	100.47	214.19	214.19
400	0.2	0.60	100.26	213.68	213.68
400	0.5	0.60	101.05	215.40	215.40

Table 9: LTE positioning and navigation results for the 100 s outage.

σ_ψ	BS	LTE only (m)		LTE-aided INS (m)			Improvement (%)	
(deg)	count	RMSE	Maximum	RMSE	Maximum	Final	RMSE	Final
0.1	4	51.55	232.99	153.56	299.47	299.16	46.54	55.59
0.1	8	60.55	219.92	38.46	67.47	61.06	86.61	90.93
0.2	4	103.65	473.03	188.27	381.27	381.07	33.76	42.91
0.2	8	107.65	431.35	23.66	41.19	40.61	91.68	93.92
0.5	4	263.06	1194.68	330.55	734.52	734.52	-25.40	-18.34
0.5	8	248.91	1061.17	50.10	80.58	42.79	80.99	93.11

5.7 LTE Position-Fix Availability

Every generated LTE position fix passed the normalized innovation gate. The number of fixes depended only on the outage duration.

5.8 Overall Comparison

LTE aiding improved the outage RMSE in 11 of the 12 configurations. The improvement ranged from −25.40 % to 91.68 %. The eight-station configuration achieved a lower LTE-aided RMSE than the corresponding four-station configuration in all six matched comparisons.

6 Discussion

6.1 Overall Interpretation

The PSS/SSS detector and PCI-based association achieved 100 % accuracy in both the four- and eight-station configurations. Therefore, differences in navigation performance were caused by the subsequent mapping, bearing estimation, and navigation-fusion stages rather than incorrect cell association.

Across the 12 configurations, LTE aiding reduced the outage RMSE in 11 cases and increased it in one case. The RMSE improvement ranged from −25.40 % to 91.68 %. The best result was obtained with an outage starting at 100 s, a heading-noise standard deviation of 0.2°, and eight base stations.

The standalone LTE accuracy varied from 51.55 m to 388.58 m RMSE. More accurate standalone LTE fixes generally provided more useful EKF corrections. However, the effect of each fix also depended on its estimated covariance and its consistency with the predicted navigation state..

Table 10: LTE positioning and navigation results for the 400 s outage.

σ_ψ	BS	LTE only (m)		LTE-aided INS (m)			Improvement (%)	
(deg)	count	RMSE	Maximum	RMSE	Maximum	Final	RMSE	Final
0.1	4	85.20	243.37	96.61	197.74	197.63	3.85	7.73
0.1	8	64.47	225.06	85.64	165.58	165.32	14.76	22.82
0.2	4	161.14	477.02	97.04	197.13	196.87	3.22	7.87
0.2	8	125.02	445.97	84.44	158.93	158.44	15.78	25.85
0.5	4	388.58	1178.39	92.25	172.45	171.51	8.71	20.38
0.5	8	304.32	1102.43	76.44	122.30	119.33	24.35	44.60

6.2 Effect of Heading Uncertainty

Increasing the heading-noise standard deviation from 0.1° to 0.5° consistently increased the standalone LTE positioning RMSE for every matched outage-time and base-station-count combination.

Heading uncertainty affects several parts of the system. It is used to rotate the TDOA-derived body-frame directions into the east-north frame, to convert the reference-receiver position to the aircraft centre, and to correct the EKF heading state. It is also used during pre-outage base-station mapping.

A small heading error may produce a considerable cross-track position error because the base stations are several kilometres from the aircraft. Approximately,

$$e_{\text{cross}} \approx \ell\, \delta\psi, \tag{47}$$

where ℓ is the distance to the base station and $\delta\psi$ is the heading error in radians. Therefore, heading accuracy is an important limitation of bearing-based positioning.

6.3 Effect of Base-Station Count

Increasing the number of base stations from four to eight reduced the standalone LTE RMSE in four of the six matched comparisons. It reduced the LTE-aided navigation RMSE in all six comparisons.

Additional base stations provide more bearing lines and can improve the redundancy of the position solution. However, improvement is not guaranteed because the usefulness of a station depends on its bearing direction, distance, mapping accuracy, and measurement uncertainty.

If several stations produce nearly parallel bearing lines, they may provide little additional information in the poorly observed direction. Furthermore, the four-station configuration is a subset of the eight-station configuration. Consequently, changing the station count also changes the angular distribution of the available bearings. The comparison therefore measures the combined effect of station count and geometry.

6.4 Effect of Outage Starting Time

The two outage starting times produced different pre-outage conditions. For the 100 s outage, the EKF and base-station mapping algorithm had a shorter period of GNSS availability. For the 400 s outage, more time was available for state convergence before GNSS was removed.

The outage starting time also changes the aircraft's position relative to the base stations. Therefore, the available bearing geometry during the outage differs between the two cases.

Because both outages end at 1000 s, their durations are also different. The outage beginning at 100 s lasts 900 s, whereas the outage beginning at 400 s lasts 600 s. Differences between the two cases therefore reflect four combined effects:

- outage duration;
- pre-outage EKF convergence;
- pre-outage mapping geometry; and
- aircraft location relative to the base stations.

The outage-start comparison should consequently not be interpreted as measuring only the effect of aircraft location.

6.5 Measurement Uncertainty and Innovation Gating

The tested configurations use perfect simulated TDOAs, meaning that no random timing error is added to the generated measurements. Nevertheless, a TDOA uncertainty of 0.5 ns is assumed when calculating the bearing and LTE position covariances.

Position errors can still arise from heading noise, noisy GNSS measurements used during mapping, base-station mapping error, the far-field approximation, and poor bearing geometry. Perfect TDOA measurements therefore do not imply perfect LTE position estimates.

The LTE position covariance controls how strongly an accepted fix affects the EKF. Underestimating this covariance may cause the filter to place excessive confidence in an inaccurate fix. Overestimating it may prevent useful measurements from significantly correcting the INS.

All 9012 generated LTE fixes passed the innovation gate across the 12 configurations. This does not prove that every fix was accurate. In particular, accepted measurements can still worsen the navigation solution if they contain systematic errors that are not represented correctly by their covariance.

6.6 Limitations

The principal limitations of the study are:

- the aircraft motion and localization are limited to two dimensions;
- TDOAs are generated analytically rather than estimated from the IQ waveforms;
- the nominal experiments use perfect TDOA measurements;
- each base station is processed in a separate IQ buffer;
- PCI detection is performed once and reused at every epoch;
- all selected base stations are assumed continuously visible;
- receiver synchronization is assumed to be ideal;
- multipath, fading, interference, and non-line-of-sight propagation are not modelled;
- an independent heading measurement remains available throughout the GNSS outage;
- each configuration uses one fixed random-noise realization; and
- no experimental RF, hardware, or flight data are used.

7 Conclusion and Future Work

This project developed a two-dimensional proof-of-concept simulation of LTE signals-of-opportunity aided inertial navigation during a GNSS outage. The implemented system includes aircraft and sensor simulation, PSS/SSS detection, PCI-based association, pre-outage base-station mapping, TDOA-based bearing estimation, weighted bearing-line positioning, and EKF-based navigation fusion.

A total of 12 configurations were evaluated using outage starting times of 100 s and 400 s, heading-noise standard deviations of 0.1°, 0.2°, and 0.5°, and base-station counts of four and eight.

LTE aiding improved the outage RMSE in 11 of the 12 configurations. The largest improvement was 91.68 %, while the largest deterioration was 25.40 %. The best configuration used eight base stations, a heading-noise standard deviation of 0.2°, and an outage starting at 100 s. Its heading-aided INS baseline RMSE was 284.23 m, while the LTE-aided RMSE was 23.66 m.

Increasing the base-station count from four to eight improved the LTE-aided RMSE in every matched comparison. Increasing heading uncertainty consistently degraded standalone LTE positioning. However, the fused EKF result did not vary monotonically with heading noise because the assumed heading uncertainty also changed the weight assigned to the LTE measurements. The outage-start results were additionally affected by outage duration, pre-outage filter convergence, mapping observations, and base-station geometry.

The main engineering conclusion is that LTE bearing measurements can provide useful navigation information during GNSS denial, but their benefit is not automatic. Performance depends on heading accuracy, landmark-map quality, bearing geometry, and realistic estimation of measurement uncertainty. Additional base stations are useful only when they provide sufficiently accurate and geometrically diverse bearings.

Future work should:

- estimate inter-receiver TDOAs directly from synchronized LTE IQ measurements;
- simulate multiple base stations in a common received waveform;
- extend the system to three-dimensional aircraft motion;
- include path loss, fading, multipath, interference, and non-line-of-sight propagation;
- model receiver clock and synchronization errors;
- perform Monte Carlo experiments using multiple random seeds;
- improve and validate the LTE position covariance model;
- investigate direct bearing updates within the EKF; and
- validate the method using real RF and navigation data.

Overall, the project establishes a modular simulation framework for studying LTE-aided navigation during GNSS outages. It also identifies the measurement and geometric conditions that determine whether LTE signals of opportunity improve or degrade the inertial navigation solution.